

Day 4 JavaScript Assignment

Problem 1: E-Commerce Product Inventory Dashboard

Business Scenario

You are developing the inventory module for an e-commerce platform like Amazon or Flipkart. The admin wants to see inventory statistics and identify products that need restocking.

Sample Input

2D Array:

```
[
  ['Laptop',65000,15,'Electronics'],
  ['Mouse',800,0,'Accessories'],
  ['Keyboard',1500,25,'Accessories'],
  ['Monitor',12000,8,'Electronics']
]
```

Requirements

- Display all products using loops.
- Display Stock Status using ternary operator (In Stock / Out of Stock).
- Count products having stock greater than 10.
- Calculate Total Inventory Value = Price × Stock.
- Display Inventory Health: Healthy if inventory value > ₹5,00,000 OR more than 3 products are in stock; otherwise Low.

Expected Output (Example)

Laptop - In Stock

Mouse - Out of Stock

Products with stock >10: 2

Total Inventory Value: ₹1012000

Inventory Status: Healthy

Problem 2: Student Course Progress Tracker

Business Scenario

An online learning platform wants to generate a performance report after a JavaScript assessment.

Sample Input

Marks Array:

```
[92,75,64,81,55,48,96,70,89,61]
```

Requirements

- Calculate average marks.
- Find highest and lowest marks.
- Count students scoring 60 or above.

- Use ternary operator to display Pass/Fail for each student.
- Display final class summary.

Expected Output (Example)

Average: 73.1

Highest:96

Lowest:48

Certified Students:8

Student 5: Fail

Overall Performance: Good

Problem 3: Website Analytics Dashboard

Business Scenario

You are building an analytics page that summarizes website traffic.

Sample Input

1D Array of 30 daily visitor counts (example: [1200,980,1100,1500,900,...])

Requirements

- Calculate total visitors.
- Calculate average visitors.
- Find highest and lowest day.
- Count days with visitors above 1000.
- Use ternary operator to display Growing Website or Needs Improvement.

Expected Output (Example)

Total Visitors: 34250

Average:1141

Highest:1800

Lowest:760

Days >1000:21

Status: Growing Website

Problem 4: Restaurant Ordering System

Business Scenario

Develop the billing logic for an online food delivery platform.

Sample Input

2D Array:

[['Burger',250,3],['Pizza',500,2],['Pasta',300,4]]

Requirements

- Display each ordered item.
- Calculate subtotal of every item.
- Calculate grand total.
- If Grand Total $\geq$ ₹2000 AND total quantity >5, apply 15% discount.
- Display Free Delivery or Delivery Charges using ternary operator.

Expected Output (Example)

Subtotal: Burger=750

Pizza=1000

Pasta=1200

Grand Total=2950

Discount=442.5

Final Amount=2507.5

Free Delivery

Problem 5: Admin User Management Panel

Business Scenario

Create an admin dashboard to monitor website users.

Sample Input

2D Array:

```
[[['Anand','Admin',True],['Rahul','User',True],['Priya','Admin',False],['Aman','User',False]]
```

Requirements

- Display all users.
- Count active and inactive users.
- Count Admin users.
- Grant access only if Active AND Role is Admin.
- Display final dashboard summary.

Expected Output (Example)

Active Users:2

Inactive Users:2

Admins:2

Anand: Access Granted

Priya: Access Denied

Dashboard Generated Successfully